

SABARISH MUTHUMANI

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Professional Summary

- Strong understanding of backend/frontend systems, data pipelines, and cloud platforms (AWS, Azure, Docker, Kubernetes, Terraform, Ansible), with experience integrating AI/ML frameworks and APIs into production systems.
- 3 years and 9 months of work experience and specializing in Software Engineering, Data Engineering, Cloud-native Applications, and AI-powered solutions.
- Proficient in building scalable REST APIs, microservices, and automation workflows using Python, Java, C++, Node.js, Spring Boot, with deployments on Kubernetes and CI/CD pipelines.
- I bring solid foundations in Algorithms, Data Structures, Operating Systems, Computer Networks, Computer Architecture, Storage, Multi-processor systems and Software Development Lifecycles.

Education

Virginia Polytechnic Institute and State University <i>Master of Science in Computer Engineering, GPA: 3.8/4</i>	Aug. 2023 – May 2025 <i>Blacksburg, USA</i>
Amrita Vishwa Vidyapeetham <i>Bachelor of Science in Computer Science and Engineering, 7.7/10</i>	Aug. 2016 – May 2020 <i>Coimbatore, India</i>

Work Experience

Network Engineer <i>Virginia Tech</i>	May 2025 - Present (3 months) <i>United States, Remote</i>
- Developing LLM/agent-based automation for deployment using LangChain, CrewAI, and LangGraph. - Deployed LAMP-based web applications using Ansible, Apache, PostgreSQL, and LDAP/2FA; managed SSL/TLS certificates with OpenSSL, and optimized storage/file systems to ensure reliable application performance. - Deployed compute resources across 20 Linux nodes with GPU (NVIDIA H100/L4) virtualization clusters, and monitored them using Prometheus and Grafana, enabling in-house Deep Learning workloads.	

AI Analyst <i>Mouri Tech</i>	May 2024 - August 2024 <i>Blacksburg, Virginia</i>
- Developed AI RAG-based document intelligence platform using OpenAI APIs, Azure AI, LangChain, and deployed on Azure AKS with MongoDB, Redis for National Grid, UK and Black and Decker, USA. - Built data indexes for diverse data sources into Azure Blob Storage, AWS S3, and relational databases. - Created a chat interface using React.js, Next.js and Python Flask backend and managed all these services using Azure API Management gateways for an efficient cloud product.	

Data Engineer <i>Virginia Tech</i>	August 2023 - May 2024 <i>Blacksburg, Virginia</i>
- Designed a Kafka pipeline to process thousands of mixed text/image dissertation PDFs, using NLP (tokenization, lemmatization) with NLTK and OpenAI embeddings, on Rancher Kubernetes.	

Software Engineer <i>Cisco Systems, Inc.</i>	April 2022 - August 2023 (1 year 8 months) <i>Bangalore, India</i>
- Built asynchronous RESTful APIs with multithreading (ThreadPool) and locks for R10, R9, WebexCC 2.0 and Broadcom customers in Java Spring Boot and Kafka, deployed on Kubernetes, reducing sync time by 75%. - Resolved XSS vulnerabilities, Kafka lags, Camunda workflows, Angular updates for RTMS WebexCC 2.0 following OWASP principles using SonarQube (static code analysis) and improving the performance by 30%.	

Associate Blockchain <i>National Payments Corporation of India (NPCI)</i>	May 2020 - March 2022 (1 year 10 months) <i>Hyderabad, India</i>
- Automated Kubernetes cluster setup with Docker and on-boarding of more than 150 banks to a Hyperledger Fabric FinTech platform using Bash and Python scripts, reducing setup errors by 80%. - Implemented 3 ACK Verification Logic in Node.js, using RabbitMQ and Redis increasing success rate to 90%. - Boosted token generation through batch processing in Hyperledger Fabric using Golang, scaling token issuance by 150%, for CBDC (Central Bank Digital Currency, a Nation-wide Blockchain Initiative). - Collaborated with Google Pay, PhonePe and Amazon Pay for CBDC retail integration in Node.js exposing gRPC, REST APIs for wallet KYC, token mint, burn, and transfer.	

PROJECTS

Live Streaming Application For Cricket - [Git]	Dec 2024
- Built a live streaming app using Kafka, AWS S3, EC2, Lambda, API Gateway, and Apache Airflow, while performing Exploratory Data Analysis (EDA) with Pandas and Matplotlib in Python.	
Interactive RAG Chatbot-GenAI - [Git]	May 2024
- Developed an interactive chatbot using Azure AI Document Intelligence with Deepgram (speech-to-text) and Groq for real-time analysis of invoice images.	

Technical Skills

Programming and Frameworks: Python, Java, C++, Go, Node.js, Spring Boot, Flask, React.js, Next.js, Bash
Data/AI/ML: NLP (NLTK, Transformers), LangChain, Anthropic, Vector DBs, NumPy, TensorFlow, PyTorch
Cloud: AWS (EC2, S3, Lambda, API Gateway, RDS), Terraform, Jenkins, GitHub, CI/CD, Ansible, Linux/Unix